PART-A **5*2 =10**

Answer all Questions

1. Define disjunctive and conjunctive normal forms and explain with an examples.
2. Given the relation R = {(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)}, decide whether it is reflexive or symmetric or anti-symmetric or transitive
3. Let f: R→ R, g: $R \rightarrow R$, where R is the set of real numbers be given by f(x) = $x^2 - 2$ and g(x) = x+4 find fog and gof. State whether these functions are bijective or not.
4. Let A = {1,2,3,4} and R be a relation on A defined by xRy if and only if x divides y. Then draw the diagram and write a matrix on R.
5. Consider the following relations on {1, 2, 3, 4}:
   R1 = {(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)},
   R2 = {(1, 1), (1, 2), (2, 1)},
   R3 = {(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)},
   R4 = {(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)},
   R5 = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)},
   R6 = {(3, 4)}.
   Which of these relations are reflexive?

PART-B **4*5 = 20**

Answer any 4 Questions

1. Verify that, for any three propositions P,Q,R the compound propositions
   [P → (Q → R)] → [(P → Q) → (Q → R)] is a tautology or not?
2. Obtain principal disjunctive and conjunctive normal forms of the following formulas
   (a) (⌐P v ⌐ Q) → (P ↔ Q) (b) (P → (P ∧ (Q → P))
3. Prove that the relation R defined by “a is congruent to b modulo m” on the set of integers is an equivalence relation.
4. Draw the Hasse diagram representing the positive divisors of 45.
5. If R denotes a relation on the set of all ordered pairs of positive integers by (a,b)R(c,d) iff ad=bc, show that ‘R’ is an equivalence relation.
6. Show that G = {1, w, w^2 } is an abelian group under multiplication.